

Mohamad Chalhoub

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About Me

GIS and Data Analytics professional with strong experience across the ArcGIS ecosystem, combining geospatial analysis, automation, and decision-support workflows. Certified Esri Instructor with hands-on experience delivering training, technical support, and solution design for diverse stakeholders. Regularly engaged in client-facing technical consulting and pre/post-sales support, providing a holistic understanding of GIS products and operational needs. Strong background in data analytics, GeoAI, and spatial workflows applied to real-world operational contexts.

Experience

Solution and Product Analyst, Esri Lebanon – Beirut, Lebanon June 2022 – Present

- Provided end-to-end GIS support across ArcGIS Desktop, Online, and Enterprise environments, supporting operational, analytical, and decision-making needs.
- Designed and maintained geospatial workflows for data collection, processing, analysis, and visualization.
- Delivered GIS training and capacity-building sessions as a Certified Esri Instructor, supporting professionals across multiple sectors.
- Supported clients directly in pre-sales and post-sales technical engagements, advising on appropriate ArcGIS products and solutions.
- Automated geospatial and analytical workflows using Python and scripting to improve efficiency and reliability.
- Applied GeoAI and deep learning workflows for spatial analysis, including object detection, segmentation, change detection, and experimentation with GIS-LLM pipelines.

Freelance Software Developer, Remote June 2021 – June 2022

- Designed and developed custom software and automation solutions to support client operational needs.
- Built lightweight tools and scripts to improve data handling and process efficiency.

Education

Lebanese University, Master's Degree in GIS and Data Science June 2023 – Dec 2024
Faculty of Science, Hadath

Lebanese University, Bachelor's Degree in Computer Science June 2016 – June 2019
Faculty of Science, Nabatieh

Research & Publications

Efficient Adaptation of Foundation Models for Remote Sensing Tasks Jan 2025
GeoAI Group, Lebanon

- Research conducted as part of Master's thesis in collaboration with CNRS-LB.
- Studied fine-tuning foundation models for Earth observation & remote sensing applications.
- Demonstrated practical GeoAI applications supporting large-scale geospatial data analysis.

Technologies

GIS & Geospatial Analytics:

- ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise, ArcGIS Server
- ArcGIS API for Python, ArcPy, ArcGIS JavaScript SDK, Web GIS
- Spatial analysis, spatial statistics, network analysis, raster & remote sensing
- Geodatabase design, data modeling, topology, versioned editing

- GIS data collection (Survey123), visualization, cartography, dashboards
- Automation using ModelBuilder and Python
- GIS capacity building and technical training

Data, AI & GeoAI:

- Machine Learning, Deep Learning, Data Analytics
- Computer Vision, Generative AI, LLM-based GIS workflows
- Object detection, segmentation, classification, change detection
- Model optimization, transfer learning, hyperparameter tuning
- TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib

Programming & Development:

- Python, JavaScript, TypeScript, C#, Java
- React, React Native, ASP.NET
- Automation scripts and backend integrations

Databases & Systems:

- SQL, MySQL, SQLite, MongoDB, Firebase
- Microsoft SQL Server
- Windows, Linux
- Virtualization: VMware, Docker

Project & Delivery:

- Agile Scrum
- Technical support, professional services, stakeholder-facing delivery

Certifications

Professional Certifications:

- ATD Training & Facilitation Certificate 
- ArcGIS API for Python Associate 2024 
- CCNA Routing and Switching: Introduction to Networks

GIS & Geospatial Certifications:

- Geodatabase Concept: Design and Modeling
- Creating and Sharing GIS Content Using Web GIS
- Spatial Analysis with ArcGIS
- ArcGIS Enterprise on Kubernetes

ArcGIS API for Python Certifications:

- Introduction/ Accessing Data in a Portal / Administering Content / Performing Analysis / Visualizing Data

ArcGIS Survey123 Certifications:

- Basics / Creating and Publishing Surveys / Using ArcGIS Survey123 Connect

Languages

- **English:** Fluent
- **French:** B2 - Level
- **Arabic:** Mother tongue